

Decent Coaching Classes
Std. IX (CBSE) — Mind Your Math - I
Answer Key — Ch 3 & Ch 4 (Ganita Manjari)

Section A

- (1) (c) $\sqrt{5}$
- (2) (b) 1
- (3) (c) real number
- (4) (a) $(a+b)(a-b)$
- (5) (a) 2^8

Section B

- (1) Any two rational numbers between 3 and 4, e.g. 3.1 and 3.5
- (2) $(3^2)^3 = 3^6 = 729$
- (3) $1/\sqrt{5} = (1 \times \sqrt{5})/(\sqrt{5} \times \sqrt{5}) = \sqrt{5}/5$
- (4) $(5x - 2y)^2 = 25x^2 - 20xy + 4y^2$
- (5) $\sqrt{50} = 5\sqrt{2}$, $\sqrt{18} = 3\sqrt{2}$. Sum = $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$

Section C

- (1) Draw a number line and mark O (0) and A (1), so OA = 1 unit. At A, draw AB perpendicular to OA with AB = 1 unit. Then OB = $\sqrt{1^2 + 1^2} = \sqrt{2}$ (by Pythagoras theorem). At B, draw BC perpendicular to OB with BC = 1 unit. Then OC = $\sqrt{OB^2 + BC^2} = \sqrt{2 + 1} = \sqrt{3}$. With O as centre and radius OC, draw an arc cutting the number line at point P. Point P represents $\sqrt{3}$.
- (2) $(2 + \sqrt{3})(2 - \sqrt{3}) = 2^2 - (\sqrt{3})^2 = 4 - 3 = 1$
- (3) $9x^2 - 24xy + 16y^2 = (3x)^2 - 2(3x)(4y) + (4y)^2 = (3x - 4y)^2$

Section D

- (1)(a) Let $x = 0.4444...$. Then $10x = 4.4444...$. Subtracting, $10x$ minus $x = 4 \Rightarrow 9x = 4 \Rightarrow x = 4/9$
- (1)(b) $97 \times 103 = (100 - 3)(100 + 3) = 100^2 - 3^2 = 10000 - 9 = 9991$